

YZV202E Optimization for Data Science

Homework III

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Homework Rules

- You must submit your solution as a PDF file and an .ipynb (Jupyter Notebook) solution file for programming questions.
- If you would like to, you can add hand-written solutions to the report instead of writing in L^AT_EX.
- Solutions must be self-contained and include the necessary intermediate steps to find the solution.
- Do not share any text that can be submitted as a part of an assignment (discussing ideas is okay).
- You may discuss the problems at an abstract level with your classmates, but you should not **share or copy** from your classmates or the Internet. You should submit your **own and individual** homework.
- Only electronic submissions that are sent before the deadline through Ni-nova will be accepted.
- You can use Desmos or Python for plotting.
- If you have any questions about the homework, you can send an email to Abdullah Kavaklı (kavakli23@itu.edu.tr) or Erhan Biçer (bicer21@itu.edu.tr).
- Reference books: Chong, E. K. P., & Zak, S. H. *An Introduction to Optimization* (2004); Kochenderfer, M. J., & Wheeler, T. A. *Algorithms for Optimization*.

Question 1 :

Consider a city network with water reservoirs consisting of four cities: Ankara, Bursa, Çanakkale, and Denizli. Ankara and Bursa have water reservoirs out of these four cities. These cities can make water transfer to other cities when there is a drought in a city.

In the summer, the drought was felt harsh in Çanakkale and Denizli. Çanakkale needed 55 units and Denizli needed 65 units of waters from the reservoirs. The reservoir at Ankara has 90 units of water and the reservoir at Bursa has 80 units of water. Cost of the water flow per unit from Ankara to Çanakkale is $(a \times 1000)$ TL, from Ankara to Denizli is $(b \times 1000)$ TL, from Bursa to Çanakkale is $(c \times 1000)$ TL, and from Bursa to Denizli is $(d \times 1000)$ TL. Take a as the 4th from the last digit of your student number, take b as the 3rd from the last digit of your student number, take c as the 2nd from the last digit of your student number, and take d as the last digit of your student number. If any of a , b , c or d happens to be 0, take that one as 5.

1. Formulate a linear programming problem to decide the number of units sent from each reservoir to Çanakkale and Denizli so that the total cost would be minimized.
2. Convert this problem into standard form.
3. Solve this problem analytically. Show all steps.
4. Using CVXPY, validate your solution. You can check [this tutorial](#) for a linear programming example in CVXPY.
5. Analyze the effect of a on the optimal total cost values with $a \in \{1, 2, 3, \dots, 10\}$. Visualize the optimal values with respect to varying a . Discuss.

Question 2 :

Consider the following linear programming problem:

$$\begin{array}{ll}\text{maximize} & x_1 + x_2 \\ \text{subject to} & x_1 + 2x_2 \leq 3 \\ & x_1 \geq x_2 \\ & x_1, x_2 \geq 0\end{array}$$

1. Write down the corresponding dual problem. Show all steps.
2. Implement the primal and dual problems in `cvxpy`, and solve.
3. Remove the $x_1 \geq x_2$ constraint. Did removing the constraint affect the optimal value or the solution? Discuss.

Question 3 :

Consider the following optimization problem:

$$\begin{aligned} \min_{x,y} \quad & xy \\ \text{s.t.} \quad & x^2 + y^2 = a \end{aligned}$$

where a is the last digit of your student ID.

1. Sketch contour lines of the objective function. Indicate the optimal solution(s) visually. Show the gradient directions for objective and constraint at optimal solution(s).
2. Use the method of Lagrange multipliers to solve it analytically. Show all steps.
3. How a affects the optimized solution? Discuss.
4. Verify your solution with using scipy. See the **documentation** (you are allowed to set initial points from your visual interpretation of the problem).
5. What if the objective was to maximize the function? Compare the x_1 and x_2 values for minimization and maximization. Discuss the behavior of the function within that range.
6. Do optimal point(s) satisfy SONC, SOSC? Is any optimal point local minimizer?

Question 4 :

Solve the following problem:

$$\begin{aligned} \min_{x_1, x_2} \quad & x_1^2 + x_2^2 \\ \text{s.t.} \quad & x_1 - 7 \leq 0, \\ & -x_1 + \frac{x_2^2}{4} + 4 \leq 0 \end{aligned}$$

1. Sketch contour lines of the objective function. Draw the feasible region. Indicate the optimal solution(s) visually. Comment on the activity of constraints at the optimal solution.
2. Write down the KKT conditions.
3. Solve the problem analytically using KKT conditions. Identify active constraints and derive the point(s) that satisfy the KKT conditions. Check regularity. Show all steps.
4. Use **cvxpy** to verify your result.